Unit (3) Twyman's law and Experimental Trustworthiness.

(1) Misinterpretation of the statistical Results. 40.

what it Means

This refers to common, often subtle, mistakes people when interpreting metrics like p-values, statistical significance and observed differences in A/B testing.

why it's Problem

Even results that seem "statistically significant" or obvious can be misleading - without understanding the nuances, you risk making the wrong decisions.

unit (3)

(2) Confidence Intervals

(1) Simple Explanation

> A confidence interval (CI) gives a range of values that we believe contains the true effect (like conversion rate difference) with a certain level of confidence (usually 95%).

You can think of it as:

> "We're 95% sure the true lies somewhere in this range".

(2) Real-world Analogy

Measuring a Ruler's length from a factory.

Imagine you are testing new plastic rulers from a factory. You measure 100 of them.

- Average length = 30.01 cm
- But they vary a little due to manufacturing.
- You say:
  ⇒ "I'm 95% confident the true average length of all rulers is between 29.98 and 30.04 cm".

That (29.98, 30.04) is your 95% confidence interval.

3. Math Concept (Proportions Case)

For A/B Testing:

You run an experiment with:

- control conversion rate = p_1
- treatment conversion rate = p_2

You want a CI for $p_2 - p_1$

Formula (approximate 95% CI):

$$(p_2 - p_1) \pm 1.96 \times SE$$

where:

$$SE = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

4. Python Code: Visualizing CI in A/B Test.

```
import numpy as np

# sample size
n = 5000

# Conversion rate
p1 = 0.1004
p2 = 0.1024
# standard error
se = np.sqrt((p1*(1-p1))/n+(p2*(1-p2))/n)
```

difference

diff2 = p2 - p1

95% CI

ci_low = diff2 - 1.96 * se

ci_high = diff2 + 1.96 * se

print(f"Observed difference: {diff2:.4f}")

print(f"95% confidence interval: ({ci_low:.4f}, {ci_high:.4f})")

output

observed difference: 0.0020

95% CI: (0.0002, 0.0042).

Interpretation

- Treatment CR was 0.2% higher.
- But the CI goes from -0.02% to +0.42%.
- That means:

> we're 95% confident the true effect is somewhere between a slight drop and a moderate gain.

! Cautions

If your CI includes 0, then:

> you cannot confidently say there's a real difference.

5) Useful Tricks to Remembers.

Concept	Meaning.
CI does not say "There's a 95% chance the true value is here.	CI means: "If we ran the experiment 100 times, 95 of them would contain the true value.
Narrower CI = More Precision	Comes from large sample size or less variance.
If CI excludes 0 → Statistically significant.	Strong evidence treatment ≠ control.
CI helps gauge practical significance	Not just "is it different?" but "how much different?".

6) Contrast to P-value.

P-value	Confidence Interval.
Tells if there's evidence of an difference	Tells how much the difference might be.
Doesn't show size or direction	Shows range and direction.
often misused or misunderstood	CI gives better intuition.

4) in Summary

"Confidence intervals give you a range of plausible values for the effect you're measuring — and tell you how confident you should be in that range."

(3) Treat to Internal Validity

(1) Simple Explanation:

Internal validity means:

> "Are we confident the observed difference in our experiment is caused by the treatment, and not by something else?"

A treat to internal validity happens when:

> Something other than the treatment affects the outcome — which confuses or contaminates the results

So even if your stats are perfect, the experiment design may be broken.

(2) Real-World Analogy:

"Bad Lighting in an Eye Test"

Imagine you're testing a new pair of reading glasses:

- Half the people try the new glasses.
- The other half use the old ones.

But — the room where people test the new glasses has dim lighting, while the old one is bright.

Now if the new glasses perform worse, is it:

- Because the glasses are bad?
- Or because the room was too dark?

That's a treat to internal validity:

> The result might be due to environmental bias, not the treatment.

② Common Threats to Internal Validity in A/B Tests.

Threat	What it means	Example
Instrumentation	Different code, data collections or logging between groups	logging bug in treatment code in Fleta conversion.
Selection bias.	Users weren't randomly assigned, or filtered differently	Mobile users mostly went to treatment group.
History	An External event affects one or ~~more~~ group more.	A holiday sale happens only during the treatment phase.
Maturity (Time effects)	User change over time, not due to treatment.	Users naturally convert more on weekends.
Leakage	One variant impacts the other	Shared cache, shared user sessions between A/B groups.
Implementation bugs.	Treatment wasn't deployed as expected.	Feature flag misfired - only half saw the real change.

(4) Python Analogy : Simulating A Threat.

Goal : Test a new sign-up flow.

```
import numpy as np
from scipy.stats import ttest_ind.

np.random.seed(42).

# Sample sizes.
n_control = 10000
n_treatment = 100000

# Control group: normally distributed users # avg age 30
control_age = np.random.normal(30, 5, n_control)

# Conversion is higher for younger users (simulate
  behavior).
control_conv = (np.random.rand(n_control) <
                (0.1 + 0.002 * (35 - control_age))).
                  astype(int).
treatment_conv = (np.random.rand(n_treatment)
                  < (0.1 + 0.002 * (35 - treatment_age))).
                    astype(int).

# Analyze
  cr_control = control_conv.mean()
  cr_treatment = treatment_conv.mean().

--- pvalue = ttest_ind(treatment_conv, control_
                       conv)
```

```
print(f"control CR: {cr_control:.4f}")
print(f"Treatment CR: {cr_treatment:.4f}")
print(f"p-value: {p_value:.4f}")
```

output

```
control CR: 0.1042
Treatment CR: 0.1138
p-value: 0.0000
```

looks significant, right! But it's fake - due to biased assignment, not a real treatment effect.

what This simulates:

- Younger users convert better.
- But treatment group got more young users by chance (or bug)
- So even if no product change occurred, conversion rate looks higher.

Good Experiments Avoid

- Cross-contamination between A/B groups.
- Changing logs, metrics, or UI's mid-experiment.
- Exposing one group to events the other doesn't see
- Incorrect group assignments or rollout errors.

(4) Segment Differences.

(1) Simple Definition.

Segment differences happen when an experiment has different effect across different groups of users.
- like mobile vs desktop, new vs returning, or users from different countries

So while the overall result might say "no impact", some segments might be gaining or losing significantly
— and you're missing it.

(2) Real-World Analogy.

Fast Food Experiment

Imagine McDonald's test a new spicy burger
- Overall result: Customers liked it slightly (avg rating: 3.2 stars vs 3.0 before)
- But when segmented
  - ↳ Asia region: loved it (4.2 stars).
  - ↳ USA region: Hated it (2.1 stars).

If you only look at the overall average, you miss the fact that:
- ⇒ The burger is great for one group.
- ⇒ But hurts satisfaction in another.

③ Why it Matters in A/B Testing.

You might miss:
- Hidden wins in smaller but valuable user segment.
- Dangerous drops in key segments.
- Opportunities to targets or personalize.

④ Related Concept: Simpson's Paradox.

Segment differences often overlap with Simpson's Paradox.

> A trend that appears in the overall data disappears or reverses when broken into segments.

⑤ Python Code: Segment Masking True Effects.

Let's simulate this:
- Treatment helps mobile users.
- Treatment hurts desktop users.
- But the mix of users hides this effect.

```
import numpy as np.
import pandas as pd.

np.random.seed(42)
# Samples sizes
n = 100000
# Segment split
device_type = np.random.choice(['mobile', 'desktop'],
                size=n, p=[0.3, 0.7]) # 30% mobile.
```

# Control group conversion by device
control_cr = {'mobile': 0.10, 'desktop': 0.12}.

# Treatment group effect: +3% on mobile, -3% on desktop
treatment_cr = {'mobile': 0.103, 'desktop': 0.117}.

# Assign group
group = np.random.choice(['control', 'treatment'], size=n)

# Build DataFrame
df = pd.DataFrame({'group': group, 'device': device_type,
'converted': conversion}).

# Overall lift
overall = df.groupby('group')['converted'].mean()
lift = (overall['treatment'] - overall['control'])/overall
['control'] * 100.

print(f"Overall control CR: {overall['control']} ~~{overall: .4f}~~")
~~['control']) * 100~~
print(f"Overall treatment CR: {overall['treatment']:
.4f}")
print(f"Overall Lift: {lift:.2f}%").

# Segment-wise lift
segment = df.groupby(['device', 'group'])['converted']
.mean().unstack()
segment['lift %'] = (segment['treatment'] -
segment['control'])/segment
['control'] * 100

```
print ("\n Segment Differences:")
print (segment)
```

Sample Output

Overall Control CR: 0.1156.
Overall Treatment CR: 0.1162
Overall lift: 0.52%.

Segment Differences:

group device	control	treatment	lift %
desktop	0.1202	0.1163	-3.08
mobile	0.1006	0.1032	2.59.

Interpretation

- The overall lift looks small: +0.52%.
- But
  - ↳ Mobile users improved
  - ↳ Desktop users dropped.

So if you only look at the overall, you miss.
- Treatment is hurting the majority (desktop users)
- Treatment is helping a smaller segment (mobile).

This is a Segment Difference + Simpson's Paradox combo.

(6) Danger if Ignored.

If you:
→ Deploy based on overall lift.
→ Ignore segment breakdown.

You risk:
→ Alienating large part of your user base.
→ Under-utilising segments where treatment is very effective.

(7) Overlapping Concepts.

Concepts	Connections.
Misinterpretation of Result	You misread the overall average as a win/loss.
Simpson's Paradox	Segment trend reverses overall trend
Treats to External Validity	You assume "everyone benefits" based on averages, but it's not generalizable
Healthy Skepticism	CI or p-value might look okay – but what is hiding inside the segments?

(8) Take aways

| Always break down your A/B test result by key segments (device, country, user type etc)

Otherwise, you're flying blind.

(5) ~~Simpson's Paradox~~

(5) Treats to External Validity

(1) Simple Explanation.

External validity asks:
⇒ "Do the result of your experiment generalize to the real world?"
If no, you have a threat to external validity.

(2) Internal Validity vs External Validity.

Concept	Question Answered
Internal Validity.	Did the treatment cause the outcome?
External Validity	Will this result hold outside the experiment?

(3) Why it Matters in A/B Testing.

Even if your A/B test shows a big win, you can't assume it will:
→ work for other user groups
→ work long term
→ work in different contexts (devices, countries, seasons)

So the ad fails in production where CTR is 0.3%.
That's a classic external validity problem.
The audience was not representative of your real customers.

(4) Common Threats to External Validity

Threat type	Example
Time bias	You ran the experiment only during a holiday week (unusually high or low engagement)
User bias	You ran it only on new users, but most revenue comes from loyal users.
Geographic bias	You tested in the US, but want to launch globally.
Device bias	You launched mobile-only but apply results to desktop too.
Scale-up failure	You tested at 5% of traffic, but rollout behaves differently (e.g. latency affects, infra bottlenecks)

(5) Python Analogy: A Result that doesn't Generalize.

Goal: Simulate a test on only "new users" and generalize it — wrongly — to all users.

```
import numpy as np
import pandas as pd

np.random.seed(42)
```

```
# Simulate 10,000 and 10,000 returning users
n = 10000
new_users = np.random.binomial(1, 0.12, n) # 12% CR
ret_users = np.random.binomial(1, 0.08, n) # 8% CR,

# Treatment improves new user CR by +2%, does nothing for
  returning.
new_users_t = np.random.binomial(1, 0.14, n).
ret_users_t = np.random.binomial(1, 0.08, n)

# Only test on new users (external validity threat)
control = new_users.
treatment = new_users_t.

#Now apply it to whole population (but ret_users never
  benefitted)
full_control = np.concatenate([new_users, ret_users])
full_treatment = np.concatenate([new_users_t, ret_users_t])

# Results
def summarize(group, label):
    return f"{label} Conversion Rate: {np.mean(group):.
                                        4f}"

print("[Experiment - Only new Users]")
print(summarize(control, "control"))
print(summarize(treatment, "Treatment"))
```

```
print ("\n[Production - All Users]").
print (summarize (full_control, "control")).
print (summarize (full_treatment, "treatment"))
```

output

[Experiment - Only New Users]:
Controls Conversion Rate: 0.12±2.
Treatment Conversion Rate: 0.1402.

[Production - All users]:
Controls Conversion Rate: 0.1012
Treatment Conversion Rate: 0.1120.

Interpretation

→ Experiment shows a strong uplift (+2% absolute)
→ But when applied to all users, uplift is much smaller
→ You wrongly assumed the treatment would benefit everyone

⇒ That's a threat to external validity - the test population was unrepresentative.

(6) Simpson's Paradox.

(1) Simple Explanation

⇒ Simpson's Paradox occurs when a trends appears in individual groups but disappears or reverse when the data is combined.

In A/B testing, this means.
| you may think treatment is working (or not) overall, but when you look at segments, the opposite is true.

(2) Real-world Analogy

College Admissions Bias.
Two departments, A and B.

Gender	Department A Acceptance	Department B Acceptance
Male	90%	30%
Female	85%	25%

So per-department
- Males and Females are both accepted at high or low rates.
- Slightly higher for males, but both genders are doing OK within each department.

⇒ So no obvious bias per department.

But here's the trick - look at how many apply to each department.

Gender	Applicants to A	Applicants to B.
Male	800	200
Female	200	800

- Most Male apply to the easy depart(A)
- Most females apply to the hard depart(B)

Now calculate actual accepted students

For males,
- Department A: 800 X 90% = 720 accepted
- Department B: 200 X 30% = 60 accepted
- Total = 720 + 60 = 780/1000 = 78% acceptance

For females,
- Department A: 200 X 85% = 170 accepted.
- Department B: 800 X 25% = 200 accepted.
- Total = 170 + 200 = 370/1000 = 37% acceptance

Simpson's Paradox

Even though:
- Females had similar or slightly better rates in each department.
- The overall acceptance rate is much worse for females

why?

They applied more to the hard department, which pulled their average down.

The Paradox in One Sentence:

Trends that hold each group (departments) can reverse when groups are combined (overall) due to different group sizes or weights

This is Exactly Like A/B Testing.

if:

- Treatment performs better in each segment.
- But one segment dominates in size

→ The overall result can appear worse, misleading you.

(3) Python Code: Simulating Simpson's Paradox in A/B Test

Let's simulate a case where:

- Treatment wins in both mobile and desktop.
- But loses overall due to different group sizes.

Sample Output:

Overall Results:

group
control 0.1423
treatment 0.1379
dtype: float64

Segmented Results:

group	control	treatment
device		
desktop	0.150	0.160
mobile	0.100	0.120

Interpretation

- within both segments, treatment performs better.
  - Desktop : 15% → 16%.
  - Mobile : 10% → 12%.

- But overall
  - control CR = 14.23%.
  - Treatment CR = 13.79% (looks worse)

That's Simpson's Paradox.

(4) ~~Ke~~ Take away

"Always segment your results before making decisions."

Simpson's Paradox can:
- Reverse what you believe.
- Hide real insights.
- lead to launching harmful changes.

(5) Good Practices to avoid it.

1. Always check metrics by key segment (device, geography, new/returning).
2. Never trust only the overall average
3. Build automated breakdowns in experiments dashboards.

(7) Encourage Healthy skepticism

(1) Simple Explanation

Healthy skepticism means: Don't blindly trust experiment results — questions them intelligently Even if the data says "statistically significant", ask:

"Does this make sense?"

"Could this be a bug, a fluke, or a setup issue?"

Famous Rule from the Book:

> Twyman's Law:
> "Any figure that interesting or different is usually wrong"

This law is a warning: if something looks too good to be true, it probably is,

(2) Real world Analogy.

(1) Miracle Drug Claim:

-) "Increase memory by 300%!"

-) But the study had 10 participants, all from the same lab.

would you:

- Immediately start taking it?
- Or question the design, bias, sample size, and agenda?

That's healthy skepticism.

(3) In A/B Testing — Examples where skepticism saves you

- case 1 : Huge Lift Detected
  - Treatment shows +25% conversion lift.
  - But that's way above expectation
  - A healthy skeptic asks.

  ↳ Did we check logging? Instrument bugs? Segment imbalances?"

- case 2 : Tiny change, Big Effects
  - Button color changes show huge revenue gain.
  - Skeptic asks:

    ↳ "Was this test running during a holiday?"

    ↳ "Is it real or a confounding variable?".

(2) Linked Topics.

Concept	Why it Overlaps
Mis interpretation of Results	skepticism helps prevent overlooking false signals.
Threats to Internal validity.	Ask: "could a bug a confounder explain this?"
Simpson's Paradox	Ask: "Did we check segments?
External Validity	Ask: "will this generalize to real user / time / countries?"

(4) Python Analogy : Unexpected Result Triggering skepticism.

Let's simulate a suspiciously huge lift in an A/B test.

Output

Suspicious

Control CR : 0.1000

Treatment CR : 0.2002

Lift : 100.20%

Skeptic's Response

"Wait, this treatment doubled conversion? That's very unlikely for a small UI change. Did we check:

- if both groups were equal in composition?
- if logs recorded conversions the same way?
- if data lack or cache bug inflated numbers?

(5) Take aways (As a skeptic).

Question a Skeptic should Ask

"Is the effect size too big to be believable?"

"Did we check for bugs, logs, or leaks?"

"Does this hold across segments?"

"Is it consistent with past experiments?"

"Will it generalize beyond this test?"

(6) Final Mental Model.

Think like a scientist, not a gambler.
Don't bet your product on pretty numbers alone.
Always verify before you trust.

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